Mechanism-anchored geometric rescoring on fungal CYP51: broad enrichment is robust, top-rank enrichment is not, and property-matched decoys explain why

Draft manuscript — OpenAFR / The Antifungal Resistance Project. Assembled 2026-08-11 from five pre-registered runs in this repository. Every number below is traceable to a `work/RESULTS_*.md` file graded against a hash-frozen pre-registration; nothing here is a new analysis, and no result was re-graded for this writeup. (§4.4 and §6.9 carry one later, clearly-labeled addition — a 2026-08-22 compound-curation finding that bounds what the method can claim about resistance; it is not a docking run and re-grades nothing.)

> **UPDATE 2026-08-23 — look #6, on an independent dataset: the decoy ceiling is REAL, not an artifact of decoy construction.** §4.1 below names experimentally confirmed inactives as "the most useful next contribution anyone could make to this benchmark", and §5's stopping rule permits exactly one thing: a new decoy construction. Both have now been done (RESULTS_verified_inactives.md, pre-registered and graded). The 7 held-out azoles were re-ranked, under the identical protocol, criterion and seed, against **279 compounds MEASURED not to inhibit *C. albicans*** (ChEMBL, consistent-inactivity rule). Result: **AUC 0.716 — the gate passes**, so the separation reported below is *not* a product of how the decoys were built, and Limitation 2 is discharged. But the tip is worse, not better: **7 measured inactives outrank the best real drug**, and of the 39 molecules inside the validated iron-bound band, **33 are compounds measured not to work and only 2 are actives**. A pre-specified subgroup split locates the effect precisely — AUC **0.810** against non-azole chemotypes versus **0.650** against measured-inactive *azole analogues*, below the bar. What the criterion mostly measures is chemotype, not potency. §4.1's bind is therefore a fact about the chemistry rather than about our decoy generator, and §4.4's "trustworthy top-~5% shortlist" should be read with the added caveat that such a shortlist will be crowded with azole-like molecules that have already been tried and failed. *(The PDF in this directory predates this update.)*

Abstract

Docking scoring functions rank known azole antifungals against fungal CYP51 **worse than random**. A mechanism-anchored geometric criterion — the minimum distance from a ligand nitrogen to the heme iron — ranks the same poses at AUC 0.794 on a pre-registered, blind held-out set (permutation $p = 0.0028$), and reproduces this on a 2,776-compound unlabeled library by pulling seven known azoles into the top 3.2%. That part holds up.

The same single-search run reported EF@1% = 12.68x. It does not survive replication. Under five-seed sampling, EF@1% collapses to 0.00x under both a best-pose (min) and a pool-size-neutral (median) aggregator, and again under an orthogonal *axial-direction* feature — while AUC *improves* to 0.862. We show the cause is not estimator noise and not the choice of geometric feature: **property-matched decoys coordinate the heme iron both closely (2.69–2.84 Å) and on-axis (2.7°–7.5°), indistinguishably from real azoles**, so six decoys outrank the best true active. We report two further findings: (i) transfer to a *C. auris* Y132F resistance pocket preserves broad separation (AUC 0.805) while eliminating top-rank enrichment via a sub-0.1 Å reshuffle; and (ii) at typical held-out sample sizes (7 actives, 355 molecules) EF@1% is arithmetically a **Bernoulli variable** taking only the values 0.00x or 12.68x — it has no resolution and should not be used as a pass/fail criterion, including by us.

We conclude that mechanism-based rescoring buys *broad* enrichment — a trustworthy top-~5% shortlist — and that a mechanism-matched decoy set structurally bounds what any mechanism-based criterion can achieve at the extreme top. All five pre-registrations, including the three that failed, are published with their hashes.

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1. Background

Structure-based virtual screening rests on a scoring function whose correlation with real affinity is, on most targets, weak. A standard response is to rescore poses against a *mechanistic* criterion instead — a pharmacophore, a metal-coordination constraint — and rank on that. The idea is decades old and is not the contribution here.

Fungal CYP51 (lanosterol 14α-demethylase) is an unusually clean test bed for it. The azole antifungals inhibit by a single, well-defined structural event: an azole-ring nitrogen coordinates the heme iron as the sixth axial ligand. The mechanism reduces to one measurable geometric quantity, which removes the usual arbitrariness in choosing a pharmacophore. The target also matters clinically: *Candida auris* is on the WHO fungal priority pathogens list and defeats azoles largely through mutations in this enzyme.

The question this work set out to answer was not “does mechanism-based rescoring beat the scoring function” — it does — but **how far does it carry, and where exactly does it stop.**

2. Methods

2.1 SYSTEM

Receptor: PDB **5TZ1**, *C. albicans* CYP51, chain A plus heme `HEM A 601`, rigid. Docking: AutoDock Vina, box 26 Å centred (70.61, 66.28, 4.18), `exhaustiveness 32`, `num_modes 20`, seed 42. Every parameter is read at both docking and grading time from a single hash-frozen `protocol.yaml` (sha256 `6eca16b1…`), whose integrity is verified by the grading script at run time, so what was docked and what was graded cannot silently diverge.

2.2 ACTIVES AND DECOYS

Actives are known azole antifungals; the primary evaluation uses **7 held out** from all criterion development (efinaconazole, luliconazole, fenticonazole, bifonazole, sertaconazole, oxiconazole, butoconazole).

Decoys were generated from ChEMBL (`scripts/make_decoys.py`), 50 per active, under three constraints chosen to defeat two specific ways this kind of benchmark cheats:

constraint	value	what it prevents
property-matched	MW ±25, cLogP ±1.5, rotatable bonds ±2, HBD ±1, HBA ±2	Docking score correlates with heavy-atom count at **r = –0.91** on this system; unmatched decoys let the pipeline “enrich” by preferring heavy molecules.
≥ 1 aromatic nitrogen	required	Without it, a nitrogen-to-iron criterion would separate perfectly while only asking *”does this molecule contain a nitrogen?”* — circular.
Morgan(r=2) Tanimoto < 0.35 vs **every** active	required	Same physical profile, different skeleton — the basis for presuming non-binding.

348 decoys survived to the graded pool (355 molecules total; 2 docking failures, both decoys). **Decoys are presumed inactive, not experimentally confirmed** — see §6.

2.3 RANKING CRITERIA

- **Mode A — Vina score** (the control).
- **Mode C — iron-approach distance:** per molecule, the minimum over poses of the distance from the nitrogen nearest the iron to the heme Fe. This is the primary criterion.
- **Mode D —** combined rank of A and C.
- **Mode E — axial (added later):** `S = |v| · (1 + sin θ)`, where `v = N − Fe` and `θ` is the acute angle between `v` and the heme normal `n = normalize((NC−NA) × (ND−NB))`, fixed once from the receptor as `(+0.6048, −0.7591, −0.2407)`. λ = 1, untuned. Rewards approaches that are both close *and* on-axis.

2.4 METRICS AND THE PRE-REGISTERED BAR

AUC, EF@1%, EF@5%, EF@10%, BEDROC(α=20), plus a 20,000-shuffle permutation test and a bootstrap CI over actives. The pass bar, declared before any held-out result was seen and never lowered, was **AUC ≥ 0.70 AND EF@1% ≥ 5.0x**, evaluated on the primary mode only.

2.5 PRE-REGISTRATION PROCEDURE

Each run has a pre-registration file committed *before* the run, hashed, and the hash verified unmodified by the grading script at grading time. Each pre-registration declares the criterion, the aggregation, the metrics, the bar, **and the conclusion to be drawn on failure** — so a negative cannot be reinterpreted after the fact. Where a run reuses existing poses, it says so and grades without new docking.

3. Results

3.1 GEOMETRY BEATS THE SCORING FUNCTION (PRE-REGISTERED, BLIND, HELD OUT)

On 7 held-out azoles among 348 property-matched decoys, sorting the *identical* poses by iron-approach distance rather than by Vina score inverts the outcome:

mode	AUC	EF@1%	EF@5%	BEDROC	role
C — iron-approach distance	**0.794**	12.68x	5.63x	0.325	primary — **PASS**
A — Vina score alone	0.471	0.00x	0.00x	0.002	control — **worse than random**
D — combined rank	0.691	0.00x	2.82x	0.126	secondary

Permutation test (20,000 label shuffles): **p = 0.0028**. Bootstrap 95% CI over actives: AUC 0.590–0.946. Held-out active ranks (of 355): 3, 8, 29, 43, 49, 124, 274.

This is the durable result. On this target, the docking program's own affinity estimate is anti-informative, and a single mechanistic distance recovers real drugs it has never seen.

A prior run (`RESULTS_validation_gate.md`) *failed* this gate — a 3.0 Å hard metal cutoff discarded 4 of 8 actives, and enlarging the box from 26 Å to 30 Å without scaling search effort degraded sampling. That failure is published alongside the pass.

3.2 THE TOP-1% ENRICHMENT DOES NOT SURVIVE REPLICATION

The run in §3.1 docked each molecule once. Re-docking the same 355 molecules under five seeds {42, 1, 2, 3, 4} and aggregating gives:

aggregator	AUC	EF@1%	EF@5%	EF@10%	BEDROC	actives in top 15
single search (seed 42)	0.794	**12.68x**	5.63x	4.23x	0.325	2/7
consensus — min over 5 seeds	0.853	**0.00x**	8.45x	5.63x	0.389	3/7
reliability — median over 5 seeds	0.830	**0.00x**	5.63x	4.23x	0.310	2/7

Every *broad* measure improves or holds under more thorough sampling. The top-1% figure goes to zero and stays there. The two aggregators bracket the plausible options — `min` takes the best pose found, `median` is pool-size-neutral and robust — and pre-registration bounded the search to exactly these two, with the “this is a ceiling, not noise” conclusion pre-committed to the failure branch. Both failed.

The `median` result is what distinguishes the two explanations. If the collapse were extreme-value bias — five searches giving decoys five chances to get lucky — a pool-size-neutral robust estimator would recover the signal. It does not. The signal is not there to recover.

3.3 IT IS NOT THE DISTANCE FEATURE EITHER: DECOYS COORDINATE AXIALLY

The one remaining explanation was that N–Fe distance is the wrong number. Azole inhibition is *axial*; a decoy might reach the iron equatorially and be a geometric impostor. We pre-registered exactly one orthogonal feature — approach direction — and graded it on the same 5-seed poses with no new docking:

ranking key	AUC	EF@1%	EF@5%	EF@10%	BEDROC
N–Fe distance, median over 5 seeds	0.830	0.00x	5.63x	4.23x	0.310
axial `S`, median over 5 seeds	**0.862**	**0.00x**	**8.45x**	**5.63x**	0.340
axial angle θ **alone**	0.861	0.00x	5.63x	—	0.240
seed-42 single-search `S`	0.819	0.00x	8.45x	—	0.320

AUC 0.862 is the best of any look taken. EF@1% is zero. Critically, adding the direction term to the *original seed-42 poses* — the ones that produced 12.68x — also gives 0.00x: the direction term pushes decoys up, not down.

The reason is visible directly in the top of the ranking:

rank	molecule	median `S`	median θ	median N–Fe (Å)	
1	decoy_CHEMBL6328_for_efinaconazole	2.881	3.4°	2.702	decoy
2	decoy_CHEMBL275096_for_oxiconazole	2.928	2.7°	2.790	decoy
3	decoy_CHEMBL10329_for_oxiconazole	3.017	3.9°	2.793	decoy
4	decoy_CHEMBL447532_for_butoconazole	3.042	7.5°	2.691	decoy
5	decoy_CHEMBL10330_for_sertaconazole	3.079	4.3°	2.842	decoy
6	decoy_CHEMBL10547_for_oxiconazole	3.094	4.3°	2.827	decoy
7	**luliconazole**	**3.133**	**6.2°**	**2.781**	**ACTIVE**

Six decoys outrank the best real drug, and they are **not off-axis impostors**: their approach angles (2.7°–7.5°) are as small as or smaller than luliconazole's (6.2°). They are doing azole-like axial iron coordination. No weighting of distance and direction can demote them, because on both axes they are doing the thing the mechanism says to do.

This is the central finding of the paper. The top-1% ceiling is not an aggregation artifact and not a missing feature. It is a statement about the pose geometry available to a rigid-receptor docking run: *closest-nitrogen distance and its approach angle saturate against a mechanism-matched decoy set.*

3.4 TRANSFER TO A RESISTANT POCKET: BROAD SEPARATION SURVIVES, THE TIP DOES NOT

The mission-relevant question is whether a wild-type-validated criterion still works once the pocket carries the dominant *C. auris* resistance substitution **Y132F**. Controlled single variable: same 7 actives, same decoys, same frozen protocol; only the residue-132 side chain changed (Tyr→Phe by deterministic deletion of the para-hydroxyl).

mode	AUC	EF@1%	EF@5%	BEDROC	role
C — iron-approach distance	**0.805**	**0.00x**	5.63x	0.235	primary — **FAIL**
A — Vina score	0.411	0.00x	0.00x	0.002	control
D — combined rank	0.677	0.00x	1.41x	0.040	secondary

AUC is essentially unchanged from wild type (0.805 vs 0.794). Actives rank 7, 16, 42, 46, 50, 79, 264 of 355. The failure is confined to the tip: **the top six slots are all decoys, reaching the iron at 2.62–2.72 Å against the best real drug's 2.722 Å.** A sub-0.1 Å reshuffle among near-tied molecules is enough to zero the metric — the same phenomenon as §3.3, now triggered by a single atom's deletion.

3.5 EXTERNAL CORROBORATION OF THE BROAD SIGNAL ON AN UNLABELED LIBRARY

Pointed at 2,776 docked compounds from the Broad Drug Repurposing Hub — screened blind, never told which are antifungals — the criterion placed seven known azoles in the top 3.2%:

flutrimazole #1 (2.371 Å), ravuconazole #9, fosfluconazole #22, fluconazole #31, voriconazole #57, luliconazole #73, croconazole #89. The heme-coordinating human CYP inhibitor **letrozole** ranked #23.

This is an embedded positive control on a genuinely novel library and it behaves consistently with §3.1–3.3: strong broad enrichment. It also exposes the per-molecule noise floor — **clotrimazole, a real azole antifungal, ranked #2774 of 2776** (best pose 17.97 Å from the iron), because docking simply failed to find a coordinating pose for it in 20 modes.

Letrozole's rank is worth stating plainly as a limitation rather than a success: the criterion measures heme coordination, and human CYPs have heme too. **Nothing in this method addresses fungal-vs-human selectivity**, which is the hard problem in antifungals specifically.

3.6 EF@1% HAS NO RESOLUTION AT THIS SAMPLE SIZE

With 7 actives among 355 molecules, the top 1% is the top 4 slots. Enrichment factor is `(a/k)/(A/N)`, so:

actives in top 4	EF@1%
0	**0.00x**
1	**12.68x**
2	25.36x

The metric is, in practice, a **Bernoulli variable**: every EF@1% we ever reported was either 0.00x or 12.68x, and the entire "12.68x → 0.00x collapse" is one molecule crossing rank 4. EF@5% is similarly quantized in steps of 2.82x (2 actives in the top 18 → 5.63x; 3 → 8.45x).

Verified directly against the implementation this project grades with (`rdkit.ML.Scoring.CalcEnrichment` via `openafr/scoring.py`), not derived analytically:

```
from openafr.scoring import metrics
N, A = 355, 7
for k in range(3):                                  # k actives in the top 4
    flags = [1]*k + [0]*(4-k) + [1]*(A-k) + [0]*(N-4-(A-k))
    print(k, metrics(flags)[1][0])                  # -> 0.0, 12.68, 25.36
```

We flag this against our own design: **the pre-registered gate should not have been conditioned on EF@1%**, because at n = 7 the metric cannot take a value between "fail" and "3.5x over the bar." This does not rescue the negative — §3.3's angle analysis establishes the ceiling mechanistically and independently of any enrichment metric — but it does mean that the *pass* in §3.1 was over-read at the time, and the subsequent "failures" were partly an instrument with two settings. The robust, resolved statements are AUC, EF@5%, and BEDROC.

4. Discussion

4.1 THE DECOY-CONSTRUCTION BIND

Benchmark decoy sets are usually criticised for being **too easy** — property-matched-but-trivially-separable decoys inflate reported performance, the well-known concern with DUD-E-style sets in machine-learning docking evaluations. Our result is the mirror image and, we think, the more instructive one.

To avoid circularity, we *required* every decoy to be a plausible iron-coordinator (≥1 aromatic nitrogen) and to match the actives on size and lipophilicity. That is the methodologically correct choice. But it means the decoy set was constructed to be indistinguishable from the actives *on precisely the axis the criterion measures*. §3.3 shows the consequence: at the tip, they are.

This generalizes into a constraint on mechanism-based rescoring:

> A mechanism-anchored criterion can separate actives from decoys only to the extent the decoys do not share the mechanism. Match decoys on the mechanistic feature, and the criterion's measured ceiling *is* the benchmark you built.

The bind is real and has no computational escape. Non-matched decoys give an inflated, circular result; mechanism-matched decoys give a criterion that saturates at the top. The only clean exit is **experimentally confirmed inactives** — compounds measured not to inhibit — rather than computationally presumed ones. We do not have those; obtaining them for CYP51 is, in our view, the most useful next contribution anyone could make to this benchmark.

4.2 PRACTICAL RECOMMENDATIONS

1. **Do not report EF@1% from a single docking run.** Ours moved from 12.68x to 0.00x on re-sampling alone, with no change to the criterion. Report multi-seed aggregates.
2. **Do not gate on EF@1% at small active counts.** Compute the achievable values first; if the metric is Bernoulli, it is not a gate (§3.6).
3. **Report AUC, EF@5%, and BEDROC together**, all under replication. On this system those were stable across four independent looks (AUC 0.79–0.86) while the tip was not.

4. **Test transfer explicitly.** A criterion validated on a wild-type structure should not be assumed to hold on the resistant variant that motivates the work (§3.4). Ours did not, at the tip.
5. **State what the mechanism does not cover.** Heme coordination does not distinguish fungal from human CYP (§3.5).
6. **Check assay precedent before proposing a shortlist for the bench.** Novel-to-a-fingerprint is not never-tested: a molecule may already have a deposited ChEMBL/PubChem record showing it inactive against fungal CYP51, and because failing measurements are under-published, a geometry rank can quietly re-run a lost experiment. We wired an exact-match plus near-neighbour precedent check as the last stage of the pipeline (`scripts/check_precedent.py`) and ran it over the top 100 of the repurposing screen (§3.5) as a worked example. It validated in-run — the only enzyme-level flags were the known azole controls (fluconazole, voriconazole), never a novel hit — and demoted seven top-ranked repurposing candidates that a whole-cell assay had already found inactive, while the near-neighbour probe against the verified-inactive azole analogues returned nothing at Tanimoto ≥ 0.85, consistent with the hits being genuinely off the azole scaffold. Absence of a record is reported as `no-precedent`, never “untested.” The full read-out is in the repository’s candidate-shortlist result.

4.3 WHAT COULD BREAK THE CEILING

Signals a rigid-receptor, single-conformer Vina pose does not contain, in rough order of promise: explicit iron–nitrogen coordination chemistry at a QM or semi-empirical level; induced-fit or ensemble receptor treatment; desolvation. None is decidable from the data used here, and we make no claim any of them would work.

4.4 WHAT THE METHOD IS GOOD FOR

Unchanged from what the data supports: **a trustworthy top-~5% shortlist for experimental triage** (AUC ~0.86, EF@5% ~8.5x), not razor-top ranking. That is a real and useful thing — it makes a wet-lab yes/no more likely than random and cheaper to reach — and it is a smaller claim than the one the first run appeared to license.

One boundary on that claim is worth stating precisely, because §3.4’s “broad separation survives Y132F” invites over-reading: **this method cannot, in principle, tell you a compound *beats* resistance.** We tried to validate resistance-awareness directly — assemble compounds with measured activity against azole-*resistant C. auris* and check whether the criterion ranks them above the azoles the mutant defeats (a 2026-08-22 curation pass, `data/ligands/PROVENANCE.md`). It cannot be done with this tool, for a structural rather than a data-gathering reason. The compounds that retain activity against the resistant strain are all large, long-tailed molecules (VT-1598, 43 heavy atoms; opelconazole, 50; the oteseconazole/posaconazole class) — over the rigid receptor's declared ≤45 heavy-atom applicability domain (§6.4), because they need the induced fit a single rigid conformer cannot provide. The small molecules the tool docks reliably are exactly the ones resistance defeats. So **resistance-retention is confounded with over-domain molecular size**, and re-discovering Y132F-defeated azoles on the mutant pocket (§3.4) demonstrates robustness of *broad enrichment*, not resistance-breaker discovery. The honest ceiling is broad azole-like enrichment on a wild-type-like pocket. Lifting it would need a flexible-receptor / induced-fit protocol that widens the domain — listed in §4.3, not claimed here.

5. Pre-registration and multiple-comparison ledger

Every look ever taken at the held-out data, in order, with its committed outcome:

#	run	question	pre-reg sha256	outcome
0	`RESULTS_validation_gate.md`	initial gate, 3.0 Å hard cutoff	—	**FAIL** (published; blocked the screen)
1	`RESULTS_run2_holdout.md`	iron-approach distance, blind holdout	`7c89d78c…`	**PASS**
2	`RESULTS_auris_Y132F.md`	does it transfer to the resistant pocket?	`689de2fd…`	**FAIL**
3	`RESULTS_consensus.md`	5-seed consensus (min)	`1c6ea042…`	**FAIL**
4	`RESULTS_reliability.md`	pool-size-neutral median — noise or ceiling?	`f8aa1ff9…`	**FAIL**
5	`RESULTS_axial.md`	one orthogonal feature: approach direction	`3ff59c29…`	**FAIL**
6	`RESULTS_verified_inactives.md`	**new dataset**: same criterion vs 279 *measured* inactives	`d2926f77…`	**PASS** (AUC 0.716)

Four looks were taken at the wild-type poses (1, 3, 4, 5). Each pre-registration bounded the number of subsequent attempts and pre-committed the failure conclusion; the reliability pre-registration foreclosed further *aggregators*, and the axial pre-registration foreclosed further *features* on this dataset. **The stopping rule is now in force: no further criterion will be tried on these poses.** Any further attempt requires an independent dataset — new actives or a new decoy construction — not another feature on this one.

Look 6 is that independent dataset, and it obeys the rule: **no new criterion, no new feature, no re-aggregation** — the same mode-C distance, the same protocol hash, the same seed, the same 7 actives, re-run against a decoy set built from measurement rather than assumption. It is the only kind of look this ledger still permits, and it was pre-registered before any of its molecules were docked.

The `RESULTS_axial.md` run also included a harness check: the reliability-distance slice reproduced the previously published reliability numbers (0.830 / 0.00x / 5.63x / 4.23x / 0.310) bit-for-bit.

6. Limitations

1. **n = 7 held-out actives.** The bootstrap 95% CI lower bound (0.590) sits *below* the 0.70 pass bar; a true AUC under the bar cannot be excluded from the primary run.
2. **Decoys are presumed inactive, not verified.** Any true binder among the 348 depresses measured performance; any systematic selection bias inflates it. This is the single largest caveat and the subject of §4.1.
3. **One decoy set, four looks.** The ceiling is demonstrated *on this set*.
4. **One rigid receptor**, *C. albicans* 5TZ1, solved with the short ligand VT-1161. No induced fit. The posaconazole/itraconazole long-tail class is a known blind spot; the declared applicability domain is ≤ 45 heavy atoms.
5. **The Y132F model is a rigid single-atom deletion.** Clinical Y132F also involves conformational change, H-bond network rearrangement, and co-occurring efflux and expression changes. K143R (clade I) and F126L (clade III) were not tested.
6. **One docking program** (Vina), one conformer per molecule, SMILES protonation as supplied.
7. **The heme normal is an idealization** estimated from four atoms of a rigid porphyrin.
8. **Nothing here has been tested against a living fungus.** Every ranked molecule is a hypothesis.
9. **Resistance-retention is confounded with molecular size, so no resistance-breaker claim is possible** (2026-08-22 curation, `data/ligands/PROVENANCE.md`; see §4.4). The verifiable pool of compounds active against azole-resistant *C. auris* is dominated by molecules above the ≤45 heavy-atom applicability domain (§6.4), where the rigid receptor fails; the compounds it docks reliably are the ones resistance defeats. A validation of resistance-*breaking* is therefore not achievable with this rigid tool, and the method's ceiling is broad azole-like enrichment only.

7. Data and code availability

All code, all five pre-registrations with their hashes, the frozen `protocol.yaml`, the actives and decoy SMILES, the full 2,776-compound ranking, and every `RESULTS_*.md` are in this repository. Docking intermediates (~180 MB) are regenerated rather than versioned; each results file carries an exact reproduce recipe.

NOTE ON FRAMING

The ingredients here are not new. Metal-coordination and pharmacophore rescoring of docking poses is a decades-old idea, and we are not the first to distrust a docking score. What this repository contributes is a pre-registered, hash-frozen, blind-holdout execution of that idea on a clinically important target — including four published failures and a mechanistic explanation for them — in a field where ranked lists are routinely published with no evidence the ranking means anything. The honest negative is the result.